Lecture 15

Seismic sources

From last time…

Linearised equations of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0,$$

subject to boundary conditions:

$$\hat{n}_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

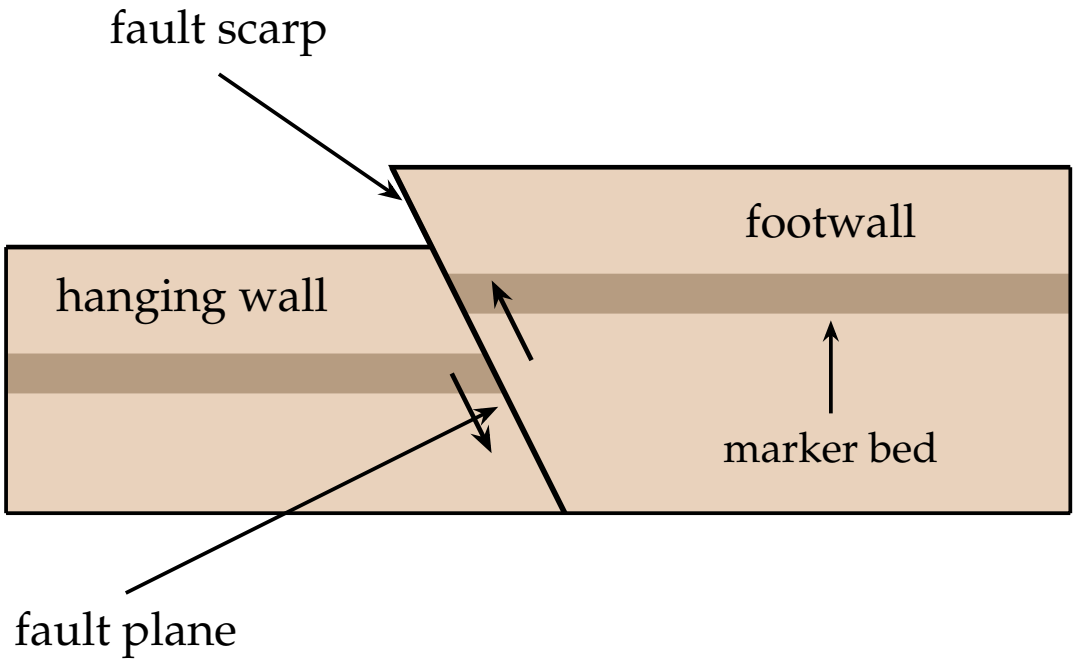
Along with initial conditions on the displacement vector and its time derivative.

But where do earthquakes enter into the theory?



The San Andreas Fault in the Carrizo Plain of California.

Ikluft, Wikimedia Commons (CC BY-SA 4.0)

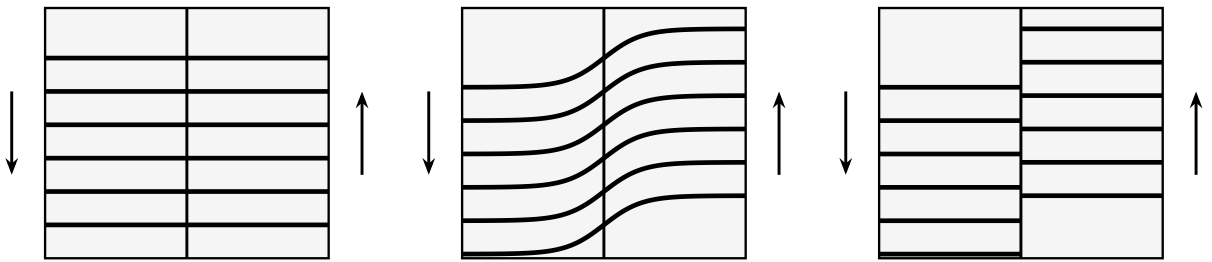


A schematic cross-section through a thrust fault. You **don't** need to learn all this terminology.



San Francisco, 1906.

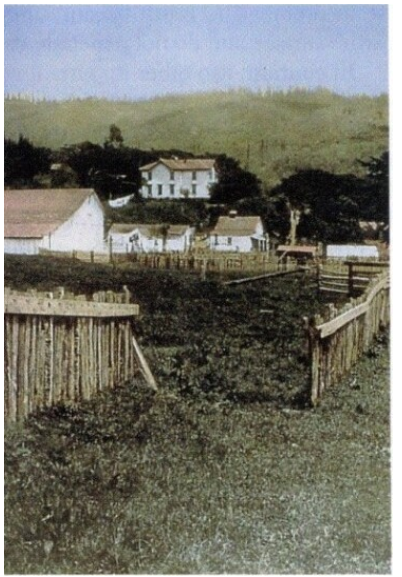
Bear Photo Co., San Francisco (1906)



(a) Rocks either side of a locked fault are driven in opposite directions

(b) The rocks deform, storing elastic energy

(c) The fault slips, the strain is released and seismic waves radiate



Elastic rebound theory of Harry Fielding Reid (1859–1944), following the 1906 San Francisco earthquake.

Photo: G. K. Gilbert, USGS (1906)

Kinematic source approximation

Lagrangian density for finite elasticity:

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F}).$$

Motion of the body satisfies **conservation of energy**:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_M E(\mathbf{x}, t)\,\mathrm{d}^3\mathbf{x} = 0,$$

$$E(\mathbf{x}, t) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}(\mathbf{x}, t)\|^2 + W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)].$$

A body in a **stable** equilibrium configuration will remain in it – no earthquakes!

To model an earthquake phenomenologically, we break the **time-translation symmetry**, adding an explicit time-dependence to the strain energy function:

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}).$$

Equations of motion are unchanged.

Constitutive equation now has an explicit time-dependence:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, t, \mathbf{F}(\mathbf{x}, t)].$$

We assume that the time-dependent strain energy function can be written:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{1}{2}\overline{S}_{ij}(\mathbf{x}, t)C_{ij}.$$

The corresponding first Piola–Kirchhoff stress tensor is:

$$T_{ij}(\mathbf{x}, t) = F_{ik}(\mathbf{x}, t)\left\{2\frac{\partial U}{\partial C_{kj}}[\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - \overline{S}_{kj}(\mathbf{x}, t)\right\}.$$

The first term on the right-hand side is the stress predicted by elasticity, but this differs from the true stress due to the second term.

The **stress glut** is defined by:

$$\begin{aligned}\overline{T}_{ij}(\mathbf{x},t) &= F_{ik}(\mathbf{x},t)\overline{S}_{kj}(\mathbf{x},t) \\ &= 2F_{ik}(\mathbf{x},t)\frac{\partial U}{\partial C_{kj}}[\mathbf{x},\mathbf{C}(\mathbf{x},t)] - T_{ij}(\mathbf{x},t).\end{aligned}$$

This term quantifies the failure of elasticity to correctly predict stresses in the body.

By definition, the stress glut is **non-zero** within the **source region**.

In the full problem, the stress glut would be determined dynamically.

Under the kinematic source approximation, the stress glut is reduced to a parameter.

We add a stress glut as a perturbation to a stress-free equilibrium configuration:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{S}{2}\overline{S}_{ij}(\mathbf{x}, t)C_{ij}.$$

Linearisation proceeds almost identically to yield:

$$\rho\frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

$$A_{ijkl}\frac{\partial u_k}{\partial x_l}\hat{n}_j = \overline{S}_{ij}\hat{n}_j.$$

Seismic source associated with an **effective body force** and **effective traction** (force per unit area) on the boundary.

Model for an ideal fault

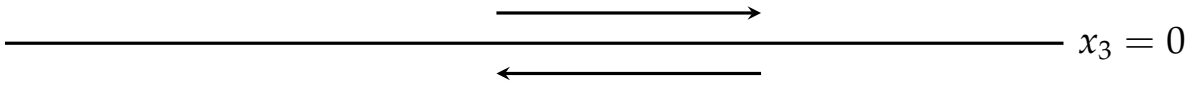


Real faults involve strong non-elastic deformation across a **finite fault zone**.



For an **ideal fault** we collapse the fault zone to a surface across which **displacement is discontinuous**.

We consider a **planar fault** and adapt our coordinates suitably:



Displacement takes the form:

$$u_i(x_1, x_2, x_3, t) = u_i^0(x_1, x_2, x_3, t) + w_i(x_1, x_2, t)H(x_3).$$

The **displacement discontinuity vector**, $\mathbf{w}$, is tangential to the fault plane ($w_3 = 0$).

Spatial derivative of the displacement vector:

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial u_i^0}{\partial x_j} + \frac{\partial w_i}{\partial x_j} H(x_3) + w_i \delta_{j3} \delta(x_3).$$

Resulting linearised stress predicted by elasticity:

$$A_{ijkl} \frac{\partial u_k^0}{\partial x_l} + A_{ijkl} \frac{\partial w_k}{\partial x_l} H(x_3) + A_{ijk3} w_k \delta(x_3).$$

Motivates the **definition** of the stress glut:

$$\overline{S}_{ij} = A_{ijk3} w_k \delta(x_3).$$

Simple generalisation to non-planar faults.

Equations of motion with an idealised fault:

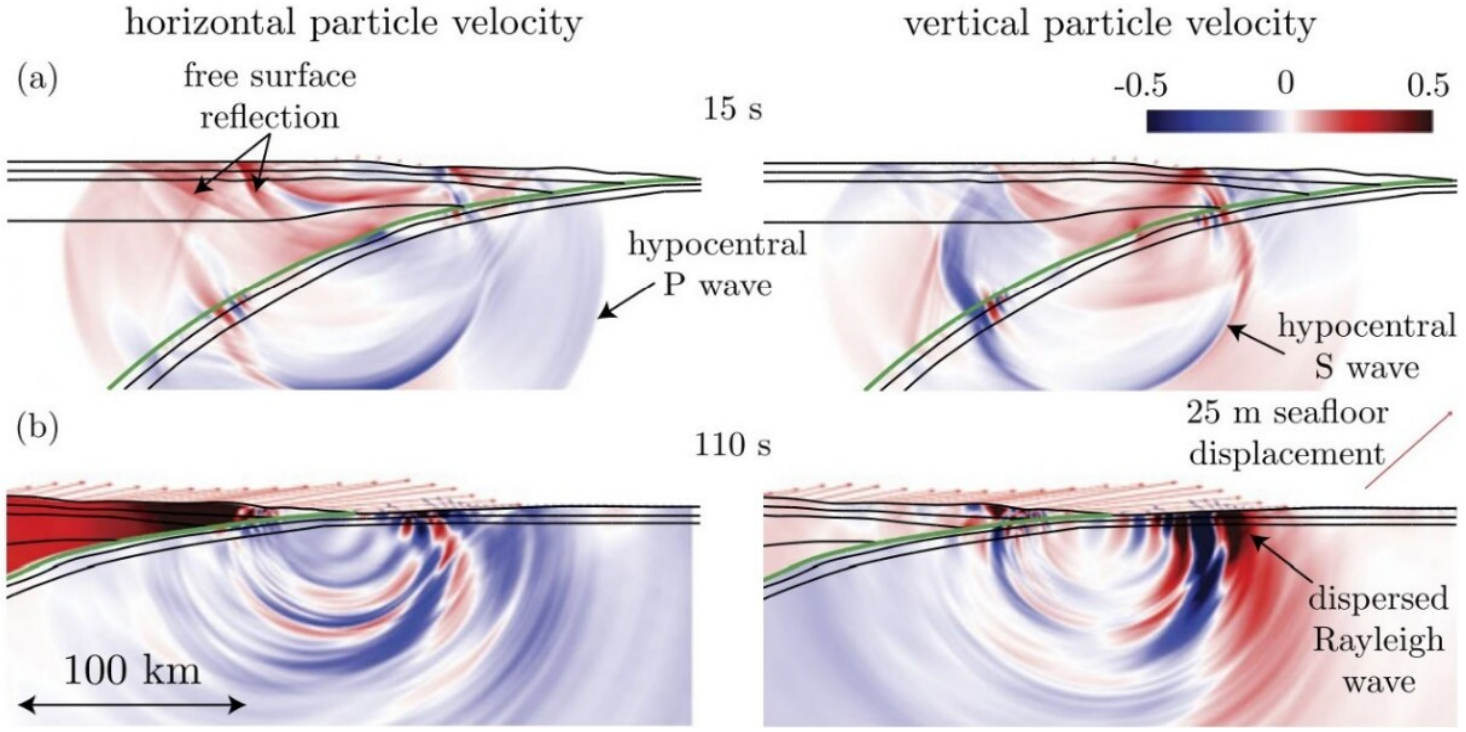
$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \overline{S}_{ij} \hat{n}_j,$$

$$\overline{S}_{ij} = A_{ijkl} w_k \hat{n}_l \delta_\Sigma.$$

Parameters in these equations:

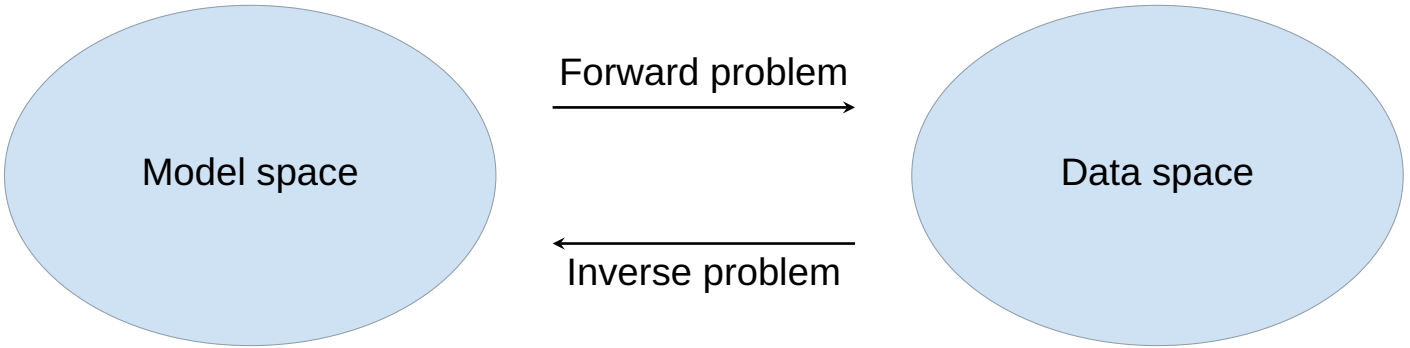
- Geometry of the earth model
- Density and the elastic tensor
- Geometry of the fault plane and the displacement discontinuity vector



Numerical simulation of seismic waves generated by rupture on a fault (2011 Tohoku earthquake).

Kozdon & Dunham (2013)

Inverse problems



This graphical representation suggests a **symmetry** between forward and inverse problems – **this is not the case!**

A mathematical problem can be viewed as a mapping between inputs and outputs.

Such a problem is said to be **well-posed** if:

1) For each input there **exists** an output
2) For each input there exists a **unique** output
3) The output depends **continuously** on the input

A problem that is not well-posed is said to be **ill-posed**.

The equations of mathematical physics are well-posed (or thought/hoped to be).

Inverse problems are almost always ill-posed.

What you need to know and be able to do

(i) Know that most natural earthquakes are generated by rapid movement across faults within the Earth.

(ii) Summarise Reid’s elastic rebound theory for earthquakes.

(iii) Explain how a seismic source can be introduced into elasticity using a stress glut, and outline the derivation of the stress glut for an idealised fault.

(iv) Understand what is meant by an inverse problem along with the associated terminology and the definition of well- and ill-posed problems.